

Table A4. POSEIDON Surface Albedo Database

Name	Database	Texture(s)	Sub-Category	SiO ₂ %	MgO %	(Min, Max) μm
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ultramafic & Mafic (SiO₂ < 63 wt%)						
Fe-oxidized ^M	H12	Powder	Hematite, Basalt	18.05 ^S	2.71 ^S	(0.30, 25.0)
Ice-rich Silicate ^M	H12	Powder	Basalt	38.47 ^S	5.78 ^S	(0.30, 25.0)
Dunite xenolith	P25	Crushed, Powder	Olivine-rich peridotite	39.04	49.28	(0.40, 25.0)
Tephrite	H25	Powder	Tephrite	41-49 ^T	N/A	(0.30, 25.05)
Clay ^M	H12	Powder	Clay	41.46 ^S	30.86 ^S	(0.30, 25.0)
Lunar anorthosite	H25	Coarse Powder	Anorthosite	44.08	0.09	(0.28, 49.86)
Basalt w/ Olivine Phenocrysts	P25	Slab, Crushed, Powder	Basalt	44.50	15.78	(0.40, 25.0)
Lunar mare basalt	H25	Coarse Powder	Basalt	44.57	11.36	(0.31, 25.99)
Feldspathic	H12	Powder	Anorthosite	44.93	0.53	(0.30, 25.0)
Harzburgite	H25	Coarse Powder	Olivine-Pyroxene rich Peridotite	<45% ^S	N/A	(0.30, 25.92)
Lherzolite	H25	Coarse Powder	Pyroxene-rich Peridotite	<45% ^S	N/A	(0.30, 25.92)
Alkaline Basalt (large)	H25	Coarse powder	Basalt	45-48 ^T	N/A	(0.30, 25.05)
Alkaline Basalt (small)	H25	Powder	Basalt	45-48 ^T	N/A	(0.30, 25.05)
Basalt glass	H25	Powder	Basalt	45-48 ^T	N/A	(0.28, 25.05)
Basalt tuff	H25	Coarse Powder	Basalt (ejecta)	45-52 ^T	N/A	(0.30, 25.05)
Gabbro	H25	Crushed	Gabbro	45-52 ^T	N/A	(0.30, 49.86)
Norite	H25	Powder	Gabbro	45-52 ^T	N/A	(0.30, 25.05)
Mars Basaltic shergottite	H25	Powder	Basalt (meteoroid)	45-52 ^T	N/A	(0.30, 49.86)
Mars breccia	H25	Slab	Basalt (impact regolith)	45-52 ^T	N/A	(0.30, 25.05)
Trachybasalt	H25	Powder	Trachybasalt	45-52 ^T	N/A	(0.30, 25.05)
EG-19-63 Olivine clinopyroxenite	P25	Slab, Crushed, Powder	Pyroxenite	47.10	24.06	(0.40, 25.0)
EG-19-70 Olivine pyroxenite	P25	Slab, Crushed, Powder	Pyroxenite	47.58	19.62	(0.40, 25.0)
Tholeiitic Basalt	H25	Slab	Basalt	48-52 ^T	N/A	(0.30, 99.72)
Basaltic andesite	P25	Slab, Crushed, Powder	Basaltic Andesite	49.24	8.53	(0.40, 25.0)
Basaltic	H12	Powder	Basalt	50.11	4.06	(0.30, 25.0)
K1919 Basalt	P25	Slab, Crushed, Powder	Basalt	50.65	6.83	(0.40, 25.0)
Ultramafic ^M	H12	Powder	Olivine, Pyroxene	50.72 ^S	49.28 ^S	(0.30, 25.0)
EG-16-68 Olivine gabbro-norite	P25	Slab, Crushed, Powder	Olivine, Gabbro	51.05	11.65	(0.40, 25.0)
Phonolite	H25	Powder	Phonolite	52.5-61 ^T	N/A	(0.30, 25.05)
STM-101 Andesite	P25	Slab, Crushed, Powder	Andesite	52.82	4.42	(0.40, 25.0)
True Intermediates						
Diorite	H25	Powder	Andesite (intrusive var.)	57-73 ^T	N/A	(0.30, 25.05)
Andesite	H25	Slab	Andesite	57-73 ^T	N/A	(0.30, 25.92)
Trachyte	H25	Powder	Trachyte	57.8-69 ^T	N/A	(0.30, 25.05)
Felsic (SiO₂ $\geq$ 63 wt%)						
Dalmatian granite	P25	Slab, Crushed, Powder	Granite	67.51	0.71	(0.40, 25.0)
Granite	H25	Powder	Granite	>69 ^T	N/A	(0.30, 25.05)
Rhyolite	H25	Powder	Rhyolite	>69 ^T	N/A	(0.30, 25.05)
Orlando gold granite	P25	Slab, Crushed, Powder	Granite	73.51	0.1	(0.40, 25.0)
Granitoid^M	H12	Powder	Granite	76.66 ^S	1.23 ^S	(0.30, 25.0)
Other						
Metal-Rich	H12	Powder	Pyrite	0 ^S	0 ^S	(0.30, 25.0)
Pyrite	H25	Powder	Pyrite	0 ^S	0 ^S	(0.12, 20.02)
Hematite (Fe-oxidized)	P25	Crushed, Powder	Hematite	2.15	0.05	(0.40, 25.0)
Hematite	H25	Powder	Hematite	0 ^S	0 ^S	(0.12, 20.02)
Albite (dust)	H25	Powder	Feldspar	68.74 ^S	0 ^S	(0.13, 20.02)
Magnesium sulfate	H25	Powder	Salt	0 ^S	33.22 ^S	(0.30, 25.92)
Extras						
White	N/A	N/A	A = 1, all μm	N/A	N/A	(0.01, 100.0)
Black	N/A	N/A	A = 0, all μm	N/A	N/A	(0.01, 100.0)
Red	N/A	N/A	A = 1, 0.620-0.750 μm	N/A	N/A	(0.01, 100.0)
Orange	N/A	N/A	A = 1, 0.590-0.620 μm	N/A	N/A	(0.01, 100.0)
Yellow	N/A	N/A	A = 1, 0.570-0.590 μm	N/A	N/A	(0.01, 100.0)

Table A4 continued

Table A4 (continued)

Name	Database	Texture(s)	Sub-Category	SiO ₂ %	MgO %	(Min, Max) μm
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Green	N/A	N/A	A = 1, 0.495-0.570 μm	N/A	N/A	(0.01, 100.0)
Blue	N/A	N/A	A = 1, 0.450-0.495 μm	N/A	N/A	(0.01, 100.0)
Purple	N/A	N/A	A = 1, 0.380-0.450 μm	N/A	N/A	(0.01, 100.0)

NOTE—The general POSEIDON v1.4 database of surface albedos (see Tables A5, A6, A7 for more specialized collections). All surface albedos present are in their directional-hemispherical reflectance form and have data in wavelengths necessary to model both surface reflection (UV and visible wavelengths) and emission (infrared wavelengths). (1) Name of the surface albedo sample in the database. ^M indicates that a sample was modeled via mineralogical mixing, otherwise a real sample was measured. We have adopted broad ultramafic + mafic ($\leq 63\%$ SiO₂), and felsic ($>63\%$ SiO₂) definitions, with the delineating percentage (63%) from Paragas et al. (2025), to further sort the table. (2) Original database the data is from; databases are Hu et al. (2012) (H12), Paragas et al. (2025) (P25) and Hammond et al. (2025) (H25). We note that H12 originally used Wyatt et al. (2001) for basalt, Cheek et al. (2009) for feldspathic, and the USGS Spectral Library (Kokaly et al. 2017) for all other data; H25 utilized the RELAB Spectral Database (Milliken et al. 2021). (3) Texture(s) of the sample measured; following the definition of Paragas et al. (2025), slab is a solid sample, crushed corresponds to coarse crushed grains (500 μm to 1mm grains) and powder corresponds to fine grains (25-80 μm) [note we extend the Paragas et al. (2025) range from 63 to 80 μm to account for many Hammond et al. (2025) samples]. We have also included coarse powders (80-500 μm). (4) Sub-category as given in Table A2. (5, 6) SiO₂ and MgO wt%, where when chemical compositional analysis was performed we report exact values, otherwise we use other methods to report a range: ^S indicates that stoichiometry was utilized to determine wt%, ^T indicates that the range of compositions was taken from the definition of the rock in the Total Alkali-Silica (TAS) Diagram (Le Maitre et al. 2005) (where identical ranges were used for intrusive and glassy variants). (7) Wavelength range of the data. Plots of albedo vs. wavelength can be seen in ‘Albedo Database’ in POSEIDON’s online documentation.

Table A5. Natural Basalt Library from First et al. (2025)

Name	Texture	Representative Geologic Process	Note	(Min, Max) μm
(1)	(2)	(3)	(4)	(5)
CNE4	Slab	Supercontinent break-up	Titanite-rich (5%)	(1.34, 27.14)
CNE6	Slab	Supercontinent break-up	Pyroxene-rich (5%)	(1.34, 27.14)
CNE9	Slab	Supercontinent break-up	Titanite-rich (5%)	(1.34, 27.14)
GOR2	Slab	Deep-mantle plume, aqueously altered	Serpentine-rich (25%)	(1.34, 27.14)
LC02	Slab	Deep-mantle plume	Fresh, unaltered basalt	(1.34, 27.14)
LC09	Slab	Deep-mantle plume	Fresh, unaltered basalt	(1.34, 27.14)
LC10	Slab	Deep-mantle plume	Fresh, unaltered basalt	(1.34, 27.14)
SE04	Slab	Mid-Ocean Ridge, aqueously altered	Amphibole-rich (50%)	(1.34, 27.14)
SE20	Slab	Mid-Ocean Ridge, aqueously altered	Amphibole-rich (59%)	(1.34, 27.14)
TABT	Slab	Volcanic-arc basalt	Alumina-rich ($\sim 20\%$)	(1.34, 27.14)
TAC9	Slab	Volcanic-arc basalt	Alumina-rich ($\sim 20\%$)	(1.34, 27.14)
TACA	Slab	Volcanic-arc basalt	Alumina-rich ($\sim 20\%$)	(1.34, 27.14)
TAPS	Slab	Volcanic-arc basalt	Alumina-rich ($\sim 20\%$)	(1.34, 27.14)
TO01	Slab	Deep-mantle plume, aqueously altered	Serpentine-rich (61%)	(1.34, 27.14)
WB04	Slab	Alkali-rich magma	Basanite	(1.34, 27.14)

NOTE—The 15 natural basaltic samples featured in First et al. (2025), in their directional-hemispherical reflectance form. Basalts are the most abundant volcanic rock on Earth, the Moon, and Mars and are expected to be the most abundant mantle rock in bulk silicate planets around FGKM stars (Putirka & Rarick 2019). This basaltic library, composed of altered and fresh basalts, can be used to test for specific mineralogical and bulk compositional signals in basalts. In particular, aqueously altered minerals (amphibole and serpentine) present in basalts have strong absorption signals that indicate the presence of past or present surface water, while bulk compositions (e.g., percentage of Al₂O₃) can probe magmatic processes. Bolded entries were featured in Figure 3 of First et al. (2025). Prefixes represent where samples were collected: CNE = Coastal New England, GOR = Gorgona, LC = Lunar Crater, SE = Santa Elena, TA = Talamanca, TO = Tortugal, WB = Wards Science. Plots of albedo vs. wavelength can be seen in ‘Albedo Database’ in POSEIDON’s online documentation.

Table A6. Lava-World Surface Library from Fortin et al. (2022)

Name (1)	Texture (2)	Parent Mantle Material (3)	Partial Melt Composition (4)	(Min, Max) μm (5)
Chond-30	Slab	Peridotite Mantle	30% Melt Fraction	(2.5, 28)
14684-30	Slab	Peridotite Mantle	30% Melt Fraction	(2.5, 28)
65639-30	Slab	Peridotite Mantle	30% Melt Fraction	(2.5, 28)
81262-30	Slab	Peridotite Mantle	30% Melt Fraction	(2.5, 28)
MM3-10	Slab	Peridotite Mantle	10% Melt Fraction	(2.5, 28)
6856-10	Slab	Peridotite Mantle	10% Melt Fraction	(2.5, 28)
6856-30	Slab	Peridotite Mantle	30% Melt Fraction	(2.5, 28)
6856-50	Slab	Peridotite Mantle	50% Melt Fraction	(2.5, 28)
59639-30	Slab	Pyroxenite Mantle	30% Melt Fraction	(2.5, 28)
22907-30	Slab	Pyroxenite Mantle	30% Melt Fraction	(2.5, 28)
MORB-10	Slab	Quartz-normative Pyroxenite Mantle	10% Melt Fraction	(2.5, 28)
MORB-30	Slab	Quartz-normative Pyroxenite Mantle	30% Melt Fraction	(2.5, 28)
MORB-50	Slab	Quartz-normative Pyroxenite Mantle	50% Melt Fraction	(2.5, 28)
MORB-100	Slab	Quartz-normative Pyroxenite Mantle	100% Melt Fraction	(2.5, 28)

NOTE—The 16 potential lava world surfaces featured in Fortin et al. (2022), which are directional-hemispherical reflectance measurements of rapidly quenched glasses. The glasses have compositions of partial melts of a parent material (i.e., for Chond-30 the glass that was measured is 100% composed of material made from the molten component of the parent material that is 30% melted). Entries are in order of decreasing SiO_2 content in the mantle composition they are made from (see Figure 1 of Fortin et al. (2022)). Includes reference Mid-Ocean Ridge Basalt (MORB), peridotite (Earth-like mantle material, MM3), and chondritic composition (Chond) measurements. Numbered lab samples are partial melts of bulk-silicate planet (BSP, bulk planet minus metallic core) compositions from the exoplanet mantle modeling of Putirka & Rarick (2019) (where primitive mantle compositions, before crustal formation, were modeled based off of stellar compositions). Peridotite compositions are olivine-rich while pyroxenite compositions are pyroxene-rich. The predicted percentage of exoplanets with each type of mantle depends on metallic core-formation mechanisms (Mercury-like, Earth-like, and Mars-like): peridotite mantles (26.8%, 78%, 95.1%), pyroxenite mantles (59.8%, 20.6%, 4.2%), and quartz-normative pyroxenite mantles (13.3%, 1.2%, 0.2%). Not represented in this sample are magnesiowüstite-normative peridotite mantles, which represents $< 0.5\%$ of predicted exoplanet mantles. See Figure 1 of Fortin et al. (2022) for parent mantle material compositions, and resultant partial melt compositions of the samples. Users should also see Fortin et al. (2024), which explored in-situ temperature-dependent emissivity measurements. Plots of albedo vs. wavelength can be seen in ‘Albedo Database’ in POSEIDON’s online documentation.

Table A7. Surface Albedos - Reflection Alone for HWO Applications

Name (1)	Description (2)	Reflectance Type (3)	(Min, Max) μm (4)
Deciduous	Leaf Piles	Bidirectional reflectance + directional-hemispherical reflectance	(0.302, 14)
Coniferous	Needle Piles	Bidirectional reflectance + directional-hemispherical reflectance	(0.302, 14)
White Fig Tree [†]	Spring Leaves	Bidirectional reflectance	(0.35, 2.5)
Basalt	Fresh basalt	Absolute reflectance	(0.30, 2.69)
Grass	Grass Mixture	Absolute reflectance	(0.26, 2.98)
Snow	Melting Snow	Absolute reflectance	(0.35, 2.50)
Ocean	Ocean Seawater	Absolute reflectance	(0.21, 2.98)
Sand	Quartz Sand	Absolute reflectance	(0.24, 2.98)
Cyanobacteria (MarSipp)	Intertidal microbial mat	Polarized Reflectance	(0.35, 1.10)

Table A7 continued

Table A7 (continued)

Name (1)	Description (2)	Reflectance Type (3)	(Min, Max) μm (4)
Cyanobacteria (Drymat)	Dry mat	Polarized Reflectance	(0.35, 1.10)
Hypersaline biota (MarGuer)	Hypersaline intertidal microbial mat	Polarized Reflectance	(0.35, 1.10)
Purple plankton (Conpurp)	Purple plankton sulfur pool	Polarized Reflectance	(0.35, 1.10)

NOTE—Data originally collected in Goodis Gordon et al. (2025) (GG25). Data sourced from the NASA JPL ECOSTRESS Spectral Library (Baldrige et al. 2009; Meerdink et al. 2016; Meerdink et al. 2019) and USGS Spectral Library Version 7 (Kokaly et al. 2017). Microbial mat samples are from Sparks et al. (2021) and are housed at NASA Ames Research Center. Unlike Table A4, albedos includes in this database are not standardized to directional-hemispherical reflectances. ECOSTRESS entries are bidirectional in short (0.3-2.5 μm) wavelengths and directional-hemispherical in longer ones. USGS datasets are taken with a myriad of spectrometers (detailed in Table 1 in Zenodo Supplementary Material (see §6.2)) that are generally calibrated to absolute reflectances. The Sparks et al. (2021) spectra are not explicitly defined, though reflectances are measured with a polarized instrument. [†]Dataset most similar to ‘Forest’ dataset highlighted in Goodis Gordon et al. (2025). For more examples of HWO-applicable datasets, see Zelakiewicz et al. (2026). Plots of albedo vs. wavelength can be seen in ‘Albedo Database’ in POSEIDON’s online documentation.

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